NON-MF GROUPS

SAUERS

We prove that not every countable group is MF in the operator norm sense. Nonapproximability was known for the unnormalized Schatten p-norms for finite p; we settle the endpoint $p = \infty$. Let $L \leq G$ and $K \trianglelefteq G$ have property (T). If K is contained in the one-sided compression defect of L, then K lies in the MF radical of G. For $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$, this defect equals H. The group H is finitely generated, simple, has property (T), and has trivial image under every homomorphism to an MF group. Consequently, $C^*_{\mathrm{r}}(H)$ is a separable stably finite C^*-algebra that is not MF. We also construct a finitely presented torsion-free property-(T) group with full MF radical whose nontrivial quotients have full MF radical.

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1. Introduction

Not every countable group is MF. The counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

Khanh–Thanh identify H with the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4], which OpenAI used to construct the first nonsofic group [OAI, Chapter 3].

A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n \colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

We use this operator norm convention throughout. The strong convergence convention additionally requires the matrix models to recover the norms of the left regular representation [GKM, Sch].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

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Here $\bigoplus_n M_{d_n}(\mathbb{C})$ is the ideal of sequences (x_n) for which $\|x_n\| \to 0$.

A separable C^*-algebra is MF if it embeds in a norm matrix corona. For a countable group G, the algebra $C^*_{\mathrm{r}}(G)$ is separable and its faithful canonical trace makes it stably finite. The group H above gives a non-MF algebra of this form.

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. By Proposition 2.1, G is MF exactly when $\mathrm{Rad}_{\mathrm{MF}}(G) = 1$, while $\mathrm{Rad}_{\mathrm{MF}}(G) = G$ implies that every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) *and $\mathfrak{D}_G(L) = G$, then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

Every operator norm asymptotic representation is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm:

$$\mathfrak{D}_G(L) \leq R_{\infty \to 2}(G).$$

Conjugation by the compressor moves the Kazhdan projection of L below itself, and stable finiteness of the corona forces equality. For a corona representation, the complement of the Kazhdan projection is a G-invariant corner. The Kazhdan inequality forces this corner to vanish when $K \leq \mathfrak{D}_G(L)$.

Many Kazhdan groups are residually finite and hence MF.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Every homomorphism from H to an MF group is trivial. Hence H is non-MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \le L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

Since H is simple and $d \neq 1$, the element d normally generates H. The containment $d \in \mathfrak{D}_H(L)$ therefore gives $\mathfrak{D}_H(L) = H$, and Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

Theorem C (a torsion-free group with full MF radical)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *and*

$$\mathrm{Rad}_{\mathrm{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The construction combines Fournier-Facio's torsion-free property-(T) group [FF, §2] with Hull's small cancellation theorem [Hu] so that the compression defect normally generates Q.

Relation to prior work. Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable C^*-algebra is NF exactly when it is nuclear and MF [BK]. Carrión–Dadarlat–Eckhardt introduced the group property used here [CDE]. The negative solution of the Connes embedding problem [JNVWY] implies the existence of separable stably finite C^*-algebras that are not MF [GH, Proposition 6.1 and Remark 6.2]; Theorem B gives one among reduced group C^*-algebras. Nonapproximability was known for the unnormalized Schatten p-norms when $1 < p < \infty$ [DGLT, LO], and Bachner–Dogon–Lubotzky settled $p = 1$ [BDL]. The case $p = \infty$ remained open. Their criterion reduces the operator norm case to normalized Hilbert–Schmidt stability [BDL, Proposition 1.5]. Under the compression relation, conjugation preserves the normalized Hilbert–Schmidt asymptotic commutant.

Leavitt introduced the algebras $L_K(1, n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then implies that H is simple [Pre]. Property (T) follows from Ershov–Jaikin-Zapirain [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression and the MF radical

2.1. **The MF radical.** The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, hence MF; conversely, every countable MF group embeds in the unitary group of a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from G to MF groups.

Proposition 2.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and G is MF if and only if $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial.*

Proof. An intersection of kernels is normal. If $\alpha\colon G \to G$ is an endomorphism and $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, then every corona homomorphism π satisfies $\pi(\alpha(x)) = (\pi \circ \alpha)(x) = 1$. Thus the radical is fully invariant.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$. If $1 \neq x \in G/R$ and g represents x, then $g \notin R$, so a homomorphism from G to an MF group separates g from 1. It kills R and induces a homomorphism separating x. Hence G/R is residually MF and therefore MF by Korchagin [Kor, Proposition 6]; this also covers the trivial quotient.

Finally, a corona embedding gives $\mathrm{Rad}_{\mathrm{MF}}(G) = 1$ when G is MF, and the quotient result gives the converse. □

2.2. **Kazhdan transport in normalized Hilbert–Schmidt norm.** For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator norm asymptotic representations of G.

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Since $x_n^* x_n \to 1$, polar correction replaces all sufficiently late x_n by unitaries at distance $o(1)$. Thus $vv^* = 1$, and the same argument applies to every matrix amplification. If equivalent projections $p \le q$ in a stably finite algebra are implemented by $w^*w = q$ and $ww^* = p$, then w is an isometry in the finite corner qAq, so $p = ww^* = q$. $\square$

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. $\square$

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))\big(\mathcal{C}_2(V,L)\big) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V,L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \le 2\,\|A - B\|\,,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\operatorname{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. By the preceding estimate, $g \mapsto \widetilde{\sigma}(g)$ is a homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. Since $uLu^{-1} \le L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.3, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.2 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\operatorname{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. ☐

Corollary 2.5. *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.4, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\Big\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \Big\| \longrightarrow 0.$$

Therefore

$$\Big\| V_n([ucu^{-1}, \ell]) - 1 \Big\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. ☐

2.3. **Normal Kazhdan subgroups and the MF radical.**

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \operatorname{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^*)$ is the coordinate restriction of $q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g, let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\left\| \widetilde{W}_n(g) - q_n U_n(g) q_n \right\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel those coordinates. Put $r_n = \operatorname{rank}(q_n)$, choose a unitary $J_n \colon \mathbb{C}^{r_n} \to q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

Theorem 2.7 (normal Kazhdan radical theorem)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T) *with $K \leq R_{\infty\to 2}(G)$. Then*

$$K \leq \operatorname{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It lies in $R_{\infty\to 2}(\Theta(G))$, because every operator norm asymptotic representation of $\Theta(G)$ pulls back along $G \to \Theta(G)$ to one of G.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

The two ranges agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K, so every element of K has trivial corona image. Lemma 2.6 gives positive integers r_n and an operator norm asymptotic representation $W_n \colon G \to \mathcal{U}(r_n)$. Let $\widehat{\Theta} \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of K under $\widehat{\Theta}$ is zero, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s)-I_{r_n})^*(W_n(s)-I_{r_n})$$

represent b. Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s)-I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n}. This contradicts $s_0 \in K \leq R_{\infty\to 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.5 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.7 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$. □

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.4.

3. The binary Leavitt group

The Leavitt relations give $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\mathrm{EL}_3(R)$ and a nontrivial $d \in \mathfrak{D}_H(L)$. Simplicity of H then gives $\mathfrak{D}_H(L) = H$.

3.1. **The self-compression.** Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{4}$$

In particular $q \neq 0$.

We use indices $0, \ldots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{5}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0 \Psi(A) s_0 = A$, so Ψ is injective. Hence Ψ restricts to an injective group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $Y = X^{-1}$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{6}$$

A direct block multiplication gives

$$\tau \,\mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{7}$$

Lemma 3.1. *The matrix τ defined in (6) belongs to* $\mathrm{EL}_{12}(R)$*.*

Proof. For $B \in M_6(R)$, the block matrices $\left(\begin{smallmatrix} I & B \\ 0 & I \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I & 0 \\ B & I \end{smallmatrix}\right)$ lie in $\mathrm{EL}_{12}(R)$: they are the products of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with $I = I_6$ therefore expresses τ as a product of six matrices of $\mathrm{EL}_{12}(R)$. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{8}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L. These matrices generate L, so (8) follows. $\square$

3.2. **Simplicity and the full MF radical.**

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \mathrm{EL}_{12}(R)$, the subgroup $N \leq \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\mathrm{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12}. Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L, so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{9}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since H is simple, the nontrivial element d normally generates H. $\square$

Proof of Theorem B. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. The identities $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ show inductively that Y generates all elementary matrices over R; the commutator identity always has a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely generated, and Proposition 3.2 gives property (T).

The containment (8), the centrality of c relative to L, and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\!\langle d \rangle\!\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

The equality $\mathrm{Rad}_{\mathrm{MF}}(H) = H$ implies that every homomorphism from H to an MF group is trivial, and hence that H is not MF.

$C^*_{\mathrm{r}}(H)$ is separable, and its faithful canonical trace makes it stably finite. If an MF embedding $e\colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contrary to the preceding conclusion. □

4. A torsion-free group with full MF radical

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. We use Hull's small cancellation theorem [Hu, Theorem 7.1] in the following form.

Theorem 4.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $t_1, \dots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q\colon G \to Q$ such that Q is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof also shows that $\ker q$ is normally generated by m elements.

Lemma 4.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

Proof. Since G is torsion-free and $N \neq 1$, the intersections $N \cap gNg^{-1}$ are infinite, so N acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1]. Torsion-freeness makes N suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $t_1, \dots, t_m$ a finite generating set of G and with the set F. Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The quotient is torsion-free by Theorem 4.1 and finitely presented because $\ker q$ is normally generated by m elements. □

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let S be a finitely presented infinite simple torsion-free group [HL]; and let H_0 be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between $1/3$ and $1/2$ [KK, Ol]. Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds U in a finitely presented torsion-free quotient P of H_0 [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently, P has property (T). By universality, P contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let E be the double HNN extension of P with stable letters u_1, u_2 and $u_i P u_i^{-1} = P_i$; it is finitely presented and torsion-free, and its Bass–Serre action makes it acylindrically hyperbolic by the tree criterion of Minasyan–Osin [MO]. Hull's common quotient theorem [Hu, Corollary 7.4] applied to E and H_0 gives a surjection $\pi\colon E \to G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of E; since E is finitely generated, the proof of that corollary produces G_0 acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of S. Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because S is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since S commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup J centralizes Γ, and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \mathrm{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and Γ has property (T) as a quotient of P.

Lemma 4.3. *Let $\rho\colon G_0 \to L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

Proof. Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, J_L centralizes Γ_L, and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As c ranges over J_L, the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and ℓ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; hence the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and S is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. □

Proof of Theorem C. Let N be the normal closure of $\pi(S)$ in G_0. It is nontrivial, so Lemma 4.2 applied to G_0, N, and $F = \{1, \pi(s)\}$ gives $q\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group Q has property (T) as a quotient of G_0.

Let $r\colon Q \to L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in L is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in L and contains a normal generating set of L, it equals L. The group L has property (T) as a quotient of Q, and $\rho(\Gamma)$ has property (T) as a quotient of Γ. Theorem A, applied to L with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\mathrm{Rad}_{\mathrm{MF}}(L) = L$. Taking r the identity gives $\mathrm{Rad}_{\mathrm{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. □

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